

PRACTICAL NO 7

Aim

- To implement the Naive Bayes' algorithm for classification.
- To train a Naive Bayes' model using the given Indian Liver Patient dataset and calculate class probabilities.
- To evaluate the accuracy of the model on test data and analyze the results.

CODE:

```
T090_PRACTICAL7_Alpy - C:\practical\AI\T090_PRACTICAL7_Alpy (3.13.0)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve, auc

df = pd.read_csv("Indian Liver Patient Dataset (ILPD).csv")

print("Dataset:")
print(df.head())
print(df.shape)

df['gender'] = df['gender'].map({
    'Male': 1,
    'Female': 0
})

df['alkphos'] = df['alkphos'].fillna(
    df['alkphos'].median()
)

df['is_patient'] = df['is_patient'].map({
    1: 0,
    2: 1
})

print("Missing values:")
print(df.isnull().sum())

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```
T090_PRACTICAL7_Alpy - C:\practical\AI\T090_PRACTICAL7_Alpy (3.13.0)
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print(df.isnull().sum())

X = df.drop('is_patient', axis=1)
y = df['is_patient']

scaler = StandardScaler()

X = scaler.fit_transform(X)

print("Standardized Data:")
print(X[:5])

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training Size:", X_train.shape)
print("Testing Size:", X_test.shape)

model = GaussianNB()

model.fit(X_train, y_train)

print("Model Trained")

predicted = model.predict(X_test)

print("Predicted:")
```

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T090_PRACTICAL7_Alpy - C:\practical\AI\T090_PRACTICAL7_Alpy (3.13.0)
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print("Confusion Matrix:")
print(confusion_matrix(y_test, predicted))

model_score = model.score(X_test, y_test)

print("Model Accuracy:")
print(model_score)

print(
    "Accuracy in percentage:",
    model_score * 100,
    "%"
)

predicted_probabilities = model.predict_proba(X_test)

print("Predicted Probabilities:")
print(predicted_probabilities)

y_prob = predicted_probabilities[:, 1]

fpr, tpr, thresholds = roc_curve(
    y_test,
    y_prob
)

roc_auc = auc(fpr, tpr)

print("ROC AUC:")
print(roc_auc)

plt.plot(
```

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```

T090_PRACTICAL7_AI.py - C:/practical/AI/T090_PRACTICAL7_AI.py (3.13.0)
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    y_test,
    y_prob
)

roc_auc = auc(fpr, tpr)

print("ROC AUC:")
print(roc_auc)

plt.plot(
    fpr,
    tpr,
    label="ROC curve (area = %0.2f)" % roc_auc
)

plt.plot(
    [0, 1],
    [0, 1],
    linestyle="--",
    label="Random"
)

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Receiver Operating Characteristic")
plt.legend(loc="lower right")

plt.show()

```

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OUTPUT:

```

IDLE Shell 3.13.0
File Edit Shell Debug Options Window Help

Python 3.13.0 (tags/v3.13.0:60403a5, Oct 7 2024, 09:38:07) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/practical/AI/T090_PRACTICAL7_AI.py =====
Dataset:
   age  gender  tot_bilirubin  ...  sgpt  alkphos  is_patient
0   65  Female             0.7  ...   3.3     0.90           1
1   62   Male            10.9  ...   3.2     0.74           1
2   62   Male             7.3  ...   3.3     0.89           1
3   58   Male             1.0  ...   3.4     1.00           1
4   72   Male             3.9  ...   2.4     0.40           1

[5 rows x 11 columns]
(583, 11)
Missing values:
age           0
gender        0
tot_bilirubin 0
direct_bilirubin 0
tot_proteins  0
albumin       0
ag_ratio      0
sgpt          0
sgot          0
alkphos       0
is_patient    0
dtype: int64
Standardized Data:
[[ 1.25209764 -1.76228085 -0.41887783 -0.49396398 -0.42671496 -0.35466541
 -0.31839333  0.29211961  0.19896867 -0.14752863]
 [ 1.06663704  0.56744644  1.22517135  1.43042334  1.68262856 -0.09159933
 -0.03433257  0.93756634  0.07315659 -0.65032257]]

```

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```
"IDLE Shell 3.13.0"
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is_patient      0
dtype: int64
Standardized Data:
[[ 1.25209764 -1.76228085 -0.41887783 -0.49396398 -0.42671496 -0.35466541
 -0.31839333  0.29211961  0.19896867 -0.14752863]
 [ 1.06663704  0.56744644  1.22517135  1.43042334  1.68262856 -0.09159933
 -0.03433257  0.93756634  0.07315659 -0.65032257]
 [ 1.06663704  0.56744644  0.6449187   0.93150811  0.82158795 -0.11352151
 -0.14518555  0.47653296  0.19896867 -0.17895325]
 [ 0.81935624  0.56744644 -0.37052344 -0.38705357 -0.44731402 -0.36562649
 -0.31146502  0.29211961  0.32478075  0.16671759]
 [ 1.68483904  0.56744644  0.0969023   0.18313526 -0.39375647 -0.29437943
 -0.17636295  0.75315299 -0.93334003 -1.71875971]]
Training Size: (466, 10)
Testing Size: (117, 10)
Model Trained
Predicted:
[[1 1 1 1 1 1 1 0 0 0 1 1 1 1 1 1 0 1 1 0 1 1 1 0 0 1 1 1 1 1 1 1 1 0
 1 1 1 0 1 0 1 0 1 0 1 1 1 0 1 1 1 0 1 0 1 1 1 1 0 1 1 1 1 0 0 1 0 1 1
 0 1 1 1 1 1 1 1 1 1 0 1 0 0 1 1 0 0 1 0 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 0
 1 1 1 1 0 1]]
Confusion Matrix:
[[31 52]
 [ 2 32]]
Model Accuracy:
0.5384615384615384
Accuracy in percentage: 53.84615384615385 %
Predicted Probabilities:
Squeezed text (117 lines)
ROC AUC:
0.7569099929128278
```

GRAPH:

